C++ Programming

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Object lifetime

- **Functions** encapsulate behavior
- **Classes** encapsulate both data and behavior

```
struct Point {
        double x, y;
        void move(double dx, double dy) { x += dx; y += dy; }
};
```

- Static (stack) objects are created, used, and destroyed automatically:

```
Point p{1, 2}; // constructor (default or user-defined)
p.move(1, -1);
```

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Object lifetime

Types of Storage Duration

Type	Created	Destroyed	Example
Automatic (stack)	At scope entry	Scope exit	Local variable
Static / global	Before main()	After main()	Global/static var
Dynamic (heap)	new	delete	Manual allocation

Object lifetime

RAII = Resource Acquisition Is Initialization

- Acquire a resource in a constructor
- Release it in the destructor
- Ensures automatic cleanup even on exceptions

The object’s lifetime defines the resource’s lifetime.

Stack vs Heap

```
Resource a; // stack object
Resource* b = new Resource(); // heap object
// automatic cleanup only for stack
```

- Stack objects destroyed automatically
- Heap objects require manual or smart pointer cleanup.

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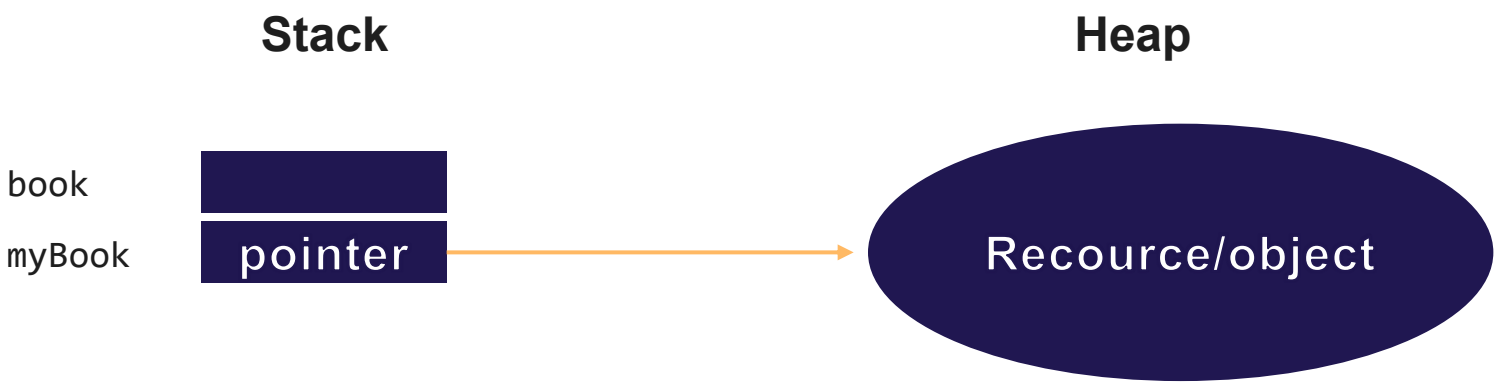
Stack and Heap

```
Book book;
Book* myBook = new Book();
```



Smart Pointers

- Smart pointers are RAII wrappers around raw pointers
- They manage heap memory automatically

```
void foo() {
        auto p = std::make_unique<Resource>();
} // destructor of unique_ptr calls delete
```

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Binary numbers

0000 0001 = 1
0000 0010 = 2
0000 0100 = 4
0000 1000 = 8
0001 0000 = 16
0010 0000 = 32
0100 0000 = 64
1000 0000 = 128

0000 0011 = 3
0000 0101 = 5
0000 0110 = 6
0000 0111 = 7
1111 1111 = 255

Bitwise operators

- & (bitwise AND) result is 1 only if both bits are 1
- | (bitwise OR) result is 1 if any of the two bits is 1
- ^ (bitwise XOR) result is 1 if two bits are different
- ~ (bitwise NOT) inverts all the bits of the number
- >> (right shift) shifts bits right by given number of steps
- << (left shift) shifts bits left by given number of steps

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